

1. Figures shown in the right margin indicate full marks
2. Answer Six questions, taking Three from each section including Q.1, Q.2, Q.5 and Q.6
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

- Q.1 (a) Mention the characteristics of free body diagram. The main arm of a crane shown in Figure Q. 1(a) has a mass of 400 kg and supports a 200 kg load and a 600 kg counterweight. The arm is connected to the vertical support via a pin joint and two flexible cables. Draw a free body diagram of the arm. [CLO1, PLO1, C3] 3

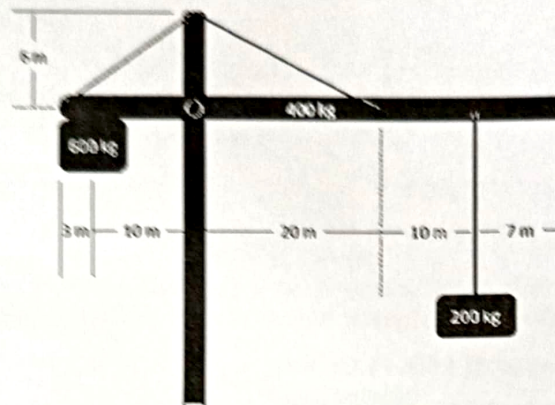


Figure Q. 1(a)

- (b) Find the magnitude and direction of the resultant force on the ring shown in Figure Q. 1(b) due to the three applied forces on it. [CLO1, PLO1, C2] 5

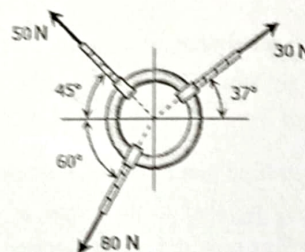


Figure Q. 1(b)

- (c) The ball D has a mass of 20 kg. If a force of $F = 100$ N is applied horizontally to the ring at A as shown in Figure: Q.1 (c), determine the largest dimension d , if the force in cable AC is zero. [CLO1, PLO1, C3] 4

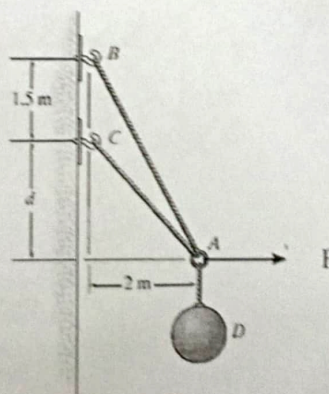


Figure Q. 1(c)

- Q.2 (a) Calculate the support reactions for the following structure shown in Figure Q. 2(a). [CLO2, PLO2, C3]

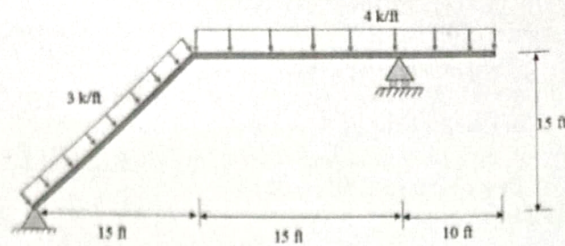


Figure Q. 2(a)

- (b) Calculate the support reactions for the beam shown in Figure Q. 2(b). [CLO2, PLO2, C3]

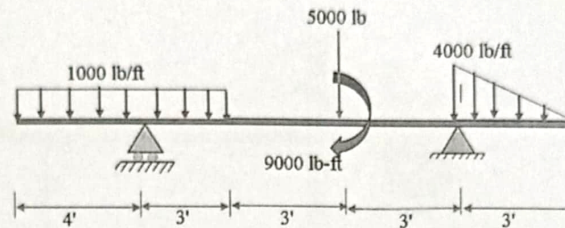


Figure Q. 2(b).

- Q.3 (a) For a parabolic chords, prove that, $F' = \sqrt{Q^2 + \left(\frac{wL}{2}\right)^2}$; where symbols bear their usual meanings. [CLO1, PLO1, C2] 5
- (b) A cable is suspended with its ends at the same elevation and 200 ft apart. The load is uniformly distributed horizontally. When the sag is 8% of the span, the maximum tension is 2040 lb. Calculate the load in pounds per ft. [CLO1, PLO1, C2] 6
- Q.4 (a) Distinguish between static and dynamic friction. [CLO1, PLO1, C2] 3
- (b) A particle moves with rectilinear motion so that its displacement is given by $S = (t^2 - 2t + 9)^{1/2}$ ft, where t is in seconds. Compute the velocity and its acceleration at the end of 3 sec. [CLO1, PLO1, C2] 3
- (c) An effort of 200 N is required just to move a certain body up an inclined plane 15° where the force is acting parallel to the plane. If the angle of inclination of the plane is made 20° the effort required, again applied parallel to the plane and found to be 230 N. Find the weight of the body and the coefficient of friction. [CLO1, PLO1, C2] 5

Section-B

- Q.5 (a) Compute the centroid of the composite area as shown in Figure: Q.5 (a). [CLO3, PLO2, C3] 6

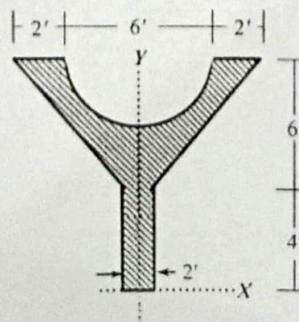


Figure: Q.5 (a)

Sketch the centroid of the shaded area as shown in **Figure: Q.5 (b)**.
[CLO3, PLO2, C3]

6

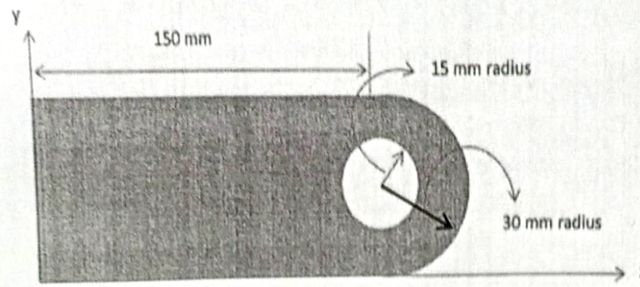


Figure: Q.5 (b)

- Q.6 (a)** Calculate the moment of inertia of the section as shown in **Figure: Q.6 (a)** about the axes passing through its center of gravity. [CLO3, PLO2, C3]

6

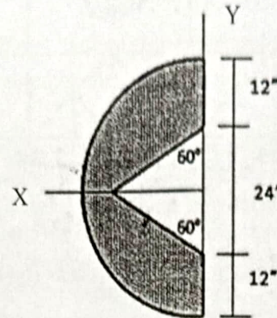


Figure: Q.6 (a)

- (b)** For the area as shown in **Figure: Q.6 (b)**, determine the moment of inertia about the axes passing through the center of gravity. [CLO3, PLO2, C3]

6

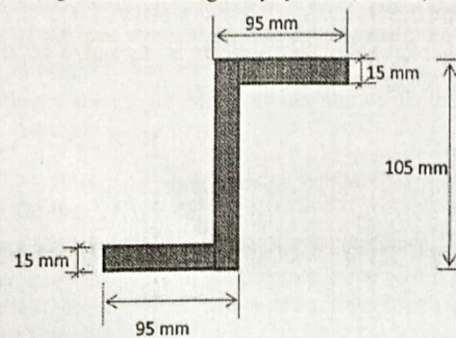


Figure: Q.6 (b)

- Q.7 (a)** Bullet A of mass 0.01 kg is moving with a velocity of 100 m/s and hit a ball B of a simple pendulum horizontally as shown in **Figure: Q.7 (a)**. [CLO2, PLO2, C3]

6

Find the maximum angle " θ " through which the pendulum swings when-

- The bullet gets embedded in the ball.
- The bullet rebounds from the surface with a velocity of 20 m/s.
- The bullet escape from the other end with a velocity of 20 m/s. Assume mass of the ball to be 1 kg and length of pendulum to be 1 meter.

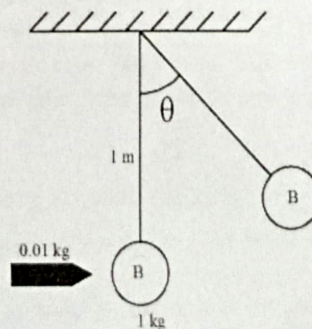


Figure: Q.7 (a)

- (b) Two discs A and B are respectively traveling on a horizontal surface with velocities 20 m/s and 8 m/s from left to right. Initially, disc B is to the right of disc A by 40 meter. The weights of A and B are 20 N and 80 N respectively.
[CLO2, PLO2, C3]

Find,

- When and where they will collide?
- If the coefficient of restitution is 0.6, Calculate their velocities after impact.
- If the impact lasts for 0.02 sec, find the average impact force.

Q.8

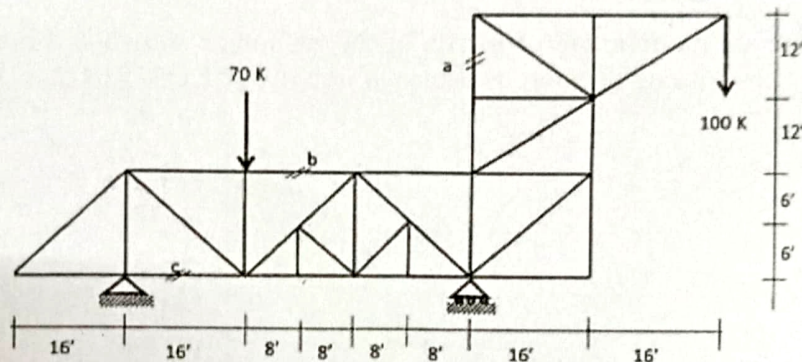


Figure: Q.8

- Determine the support reactions of the truss as shown in **Figure:Q.8**. 2
[CLO3, PLO2, C3]
- Identify zero force members of the truss as shown in **Figure:Q.8**. 4
[CLO3, PLO2, C3]
- Calculate the forces in the members a, b, and c of the truss as shown in **Figure:Q.8**. [CLO3, PLO2, C3] 5